

ORIGINAL ARTICLE

Government support and cross-border innovation: The effect of China's innovative city policy on Chinese firms' patenting in the United States

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Abstract

Firms innovate their new processes and products not only within their countries but also across national boundaries. This is known as *cross-border innovation*, and many countries have initiated policies to support it. But *how does government support influence cross-border innovation?* Through the lens of Porter's diamond model, this study conceptualizes government support as slack resources and investigates how it affects a firm's inventive activities in an uncertain, competitive global market. This study examines Chinese firms' patent filings in the United States under the enactment of the "innovative city" policy. We apply matching and generalized difference-in-differences techniques to this staggered quasi-experimental setting and find that government support stimulates firms' cross-border innovation. We further show that firms with higher research and development capacity take better advantage of government support. Finally, we show that government support leads to a rebalancing of exploratory and exploitative innovation such that a firm that has conducted more exploration will perform more exploitation and vice versa. Taken together, this study extends our understanding of slack from internal resources to government support in the context of global operations. Findings suggest that government actions that improve the ecosystem of innovation in one country strengthen firms' innovation and competitiveness overseas.

KEY WORDS

cross-border innovation, difference-in-differences, government support, policy analysis, quasi-experiment

National prosperity is created, not inherited.

— Porter (1990, p. 73)

1 | INTRODUCTION

Much of wealth creation and economic leadership comes through the innovation of new products and processes (Schilling & Shankar, 2019). Nation states may attempt to encourage firms to innovate not only within their countries but also across national boundaries. Most work on cross-border innovation focuses on firms and their foreign subsidiaries (e.g., Almeida, 1996). We define *cross-border innovation*, instead, as a firm's innovation processes and outputs across national boundaries, which include the patenting

processes that help protect the firm's innovations in a foreign country. To file a patent in a foreign country, firms, with or without foreign co-inventors, need to work with foreign attorneys and government agencies to obtain and apply knowledge (such as customer expectations and production standards) about the foreign market (Miao et al., 2021). Because patents have "value as indicators of innovation" (Gittelman, 2008; Griliches et al., 1987), the approval of foreign patents signifies a firm's successful use of foreign inputs and demonstrates its research and development (R&D) capability (Miao et al., 2021), especially in the early stages of product/process innovation (Kavadias & Ulrich, 2020).

For their part, firms increasingly engage in cross-border innovation to enhance their competitiveness. Every year from 2008 to 2019, more than half of the patents granted in the United States were filed by foreign entities with no physical presence in the country (U.S. Patent and Trademark Office,

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2021). Emerging-market firms account for a large portion of these patent assignees. Emerging markets have long been viewed as export-orientated places with cheap-labor-enabled manufacturing, and emerging-market firms as lacking “firm-specific advantages” or knowledge-based capabilities (Choo et al., 2020; Gaimon & Ramachandran, 2021). However, these firms’ increasing cross-border innovation suggests their potential to grow and compete with firms in developed countries (Y. Luo & Tung, 2007). Indeed, from 2007 to 2014, emerging-market firms’ contribution to global foreign direct investment increased from 12% to 36% (Paul & Benito, 2018).

Government support plays a critical role in promoting cross-border innovation for firms in emerging markets (Guo et al., 2016). Compared to domestic innovation, cross-border innovation has much higher levels of uncertainty (Brouthers et al., 2008; Coeurderoy & Murray, 2008; Sartor & Beamish, 2014) and “liabilities of foreignness” including “costs arising from the unfamiliarity of the environment, from cultural, political, and economic differences, and from the need for coordination across geographic distance, among other factors” (Zaheer, 1995, p. 341). Governments in emerging economies enact policies to improve the infrastructure and ecosystem for innovation, thereby enhancing emerging-market firms’ ability to innovate and compete in the global market. The policies can strengthen location advantages (Porter, 1990, 1998) and provide slack resources that facilitate product/service innovations (Hitt et al., 2016; Nohria & Gulati, 1996). This kind of policy support is common around the world (Jugend et al., 2020), especially in emerging markets (e.g., Bangladesh, Chaminade & Vang, 2008; India, Gupta & Barua, 2016).

The present study takes China, the largest emerging market, as an exemplar context of government support for cross-border innovation. China considers innovation “one of the key drivers of its future growth and convergence with more developed countries” (Medvedev et al., 2020), actively implements innovation policies, and benchmarks its success against developed countries and innovative regions such as the Silicon Valley in the United States. In this empirical setting, the central government initiated a series of policy interventions in 2009, to be implemented by targeted individual city governments; the policies intended to develop “national innovative cities” (MOST, 2010; NDRC, 2010; Shenzhen Government, 2009). Through the lens of Porter’s diamond model (Porter, 1990, 1998), this Innovative City Policy (hereafter, “the policy”) improves factors that impact a firm’s ability to compete internationally (see Supporting Information Appendix A for the key policy instruments and Supporting Information Appendix B for the policy enactment in different cities). In other words, government support implies that firms in targeted cities can benefit from a collection of complementary slack resources. For example, BYD Co., Ltd., received support from the Shenzhen government, and the policy accelerated BYD’s innovation and production worldwide (see a mini-case in Supporting Information Appendix C for more details).

However, despite the potential benefits, the actual effects of government support on firm innovation remain inconclusive in the literature (Acemoglu et al., 2018; Guo et al., 2016). Some researchers have found that government support promotes enterprise innovation (e.g., Doh & Kim, 2014; Guo et al., 2016; Hussinger, 2008), while others believe it inhibits innovation (e.g., Hong et al., 2016). In the field of operations management (OM), the interaction between government support and cross-border innovation is even less understood. Therefore, this study fills a research gap by examining, empirically, *how government support affects firms’ inventive productivity and strategy in foreign markets*. Further, we investigate what kinds of firms benefit most from government support and how these policy interventions alter firms’ innovation portfolios.

Porter’s diamond model helps understand government support, which we conceptualize as slack resources that influence firms’ inventive activities in the uncertain global market. Through generalized difference-in-differences (DiD), we compare Chinese firms’ patent applications in the United States before and after the policy shock between the “innovative cities” and the other cities without the policy intervention. The results of this natural experiment show that government support stimulates cross-border innovation and makes strong firms stronger; that is, firms supported by the policy and that had higher R&D capacity filed more U.S. patents.

We next examine a firm’s innovation portfolio in terms of both radical innovations (“revolutionary changes in technology and clear departures from existing practices”; Chao & Kavadias, 2008; Dewar & Dutton, 1986, p. 1422) and incremental innovations (“minor improvement and simple adjustments to current technologies”; Chao & Kavadias, 2008; Dewar & Dutton, 1986, p. 1423). In this, we broadly categorize innovation into *exploration* and *exploitation*, respectively (Adler et al., 2009; Benner & Tushman, 2002; March, 1991). Given that the literature on ambidexterity (Junni et al., 2013) argues that firms will seek to balance exploration and exploitation to improve long-term performance and that government support gives firms added resources for both types of innovation, we expect the policy should improve “treated” firms’ long-term performance in the global market. Indeed, we find that firms rebalance their innovation portfolios after receiving government support; that is, a firm that has conducted more exploration will perform more exploitation and vice versa. This broadly aligns with the ambidexterity literature.

This study is among the first to examine the role of government support in cross-border innovation in the multidisciplinary domain of knowledge management and management of technology (Gaimon et al., 2017). It generates a number of theoretical contributions and implications. First, this study extends our understanding of slack resources and government support through Porter’s diamond model. Rather than reducing government support to single instruments, we consider the support a collective of slack resources distributed through location advantages. As emerging-market

firms that conduct cross-border innovation are faced with riskier and more uncertain environments than those who innovate domestically, government resources can better improve firms' innovation capability and productivity in the global market. Second, this study clarifies the moderating role of a firm's R&D capacity, suggesting which firms might potentially benefit more from a policy intervention in countries with similar incentivizing policies to China's. Third, this study shows a rebalance between exploration and exploitation as a result of government support, with implications for addressing the productivity dilemma from an innovation perspective (Adler et al., 2009) and studying the allocation of different types of projects to new product development (NPD) project portfolios (Chandrasekaran et al., 2012, 2016; Kavadias & Ulrich, 2020).

The rest of this paper is organized as follows. Section 2 introduces China's policy through the lens of Porter's diamond model and conceptualizes government support as slack resources. Furthering our understanding of slack, we integrate three literature streams in OM, namely slack, absorptive capacity, and innovation (exploration–exploitation). We develop hypotheses based on the theory. Section 3 introduces our research method, including the data used in our analysis, the operationalization of the variables, and the empirical model. Section 4 presents the analyses and results, together with our robustness checks. Section 5 discusses the theoretical and managerial implications of our research findings and concludes the paper.

2 | THEORETICAL LENS AND HYPOTHESES

2.1 | Government support, Porter's diamond model, and slack resources

Michael Porter proposed the diamond model, also known as the Porter diamond theory of national advantage, to help understand the competitive advantage that regions possess due to certain factors available to them but not to others (Porter, 1990, 1998). Porter argued that any company's ability to compete in the international arena is based mainly on an interrelated set of location advantages, namely firm strategy, structure, and rivalry; related supporting industries; demand conditions; and factor conditions (Porter, 1990). Porter's diamond model provides a framework through which we might better understand the role of government support and the underlying mechanisms of policy intervention. Therefore, it has gained importance as governments in both emerging and advanced markets enact policies to support their firms' global operations and cross-border innovative capabilities.

Many countries implement policies to enable firms to innovate (see, e.g., Jugend et al., 2020, for a review). China, in particular, has prioritized innovation as a national strategy (Medvedev et al., 2020). In 2009, China's National Develop-

ment and Reform Commission (NDRC) approved the pilot construction of what was called national innovative cities in Shenzhen (Shenzhen Government, 2009). In the subsequent years, China's Ministry of Science and Technology (MOST) followed up with NDRC to co-develop the policy of "constructing national innovative cities" (see Supporting Information Appendix A for a list of key instruments in the policy). As a result, 60 other cities in China were approved to implement the policy on a rolling basis (MOST, 2010; MOST & NDRC, 2016).

Individual city governments implement the policy with the intent to improve location advantages and foster firm and regional competitiveness. Through the lens of Porter's diamond model, the policy intends to (1) promote competition and create pressure to innovate (mapped to Porter's firm strategy, structure, and rivalry), (2) enhance related supporting industries such as legal services (Porter's related supporting industries), (3) foster a larger, more dynamic consumer market that pressures firms to innovate faster and create better products (Porter's demand conditions), and (4) provide better factor conditions such as human resources, physical resources, knowledge resources, capital resources, and infrastructure (Porter's factor conditions) to support the effort. In sum, the policy intervention neatly maps onto Porter's four factors supporting the strength of cluster and regional innovation.

From firms' perspectives, the policy provides *slack resources* that promote their inventive activities "in excess of the minimum necessary to produce a given level of organizational output" (Nohria & Gulati, 1996, p. 1246). This conceptualization aligns with Zhou and Li (2021)'s findings that highlight "the accumulation of innovative resources" (p. 12) as the mechanism by which the policy influences innovation. With additional slack resources provided through the government, the policy enables firms to "increase expenditures on science & technology innovation, improve their independent innovation capabilities and empower innovative firms with independent intellectual property, core technologies, well-known brands, and international competitiveness" (Z. Li & Yang, 2021, p. 62). In general, government support acts as a "catalyst and challenger" to "encourage—or even push" firms to increase their innovation using a collective of various slack resources (Porter, 1990, p. 90).

2.2 | Government support in cross-border innovation

Innovation of new products and processes involves highly specialized, interdependent activities and "remains one of the most challenging efforts to perform and manage" (Kavadias & Hutchison-Krupat, 2020, p. 202). *Cross-border innovation* refers to a firm's innovation processes and outputs across national boundaries. Compared to domestic innovation, cross-border innovation is even more challenging. There

is substantial uncertainty amid rapid changes in technologies and market conditions (Sartor & Beamish, 2014), the regulatory hazard of globalization (Coeurderoy & Murray, 2008; Henisz & Delios, 2001), and the political and economic instabilities in the target country (Brouthers et al., 2008; Demirbag & Glaister, 2010), any of which may upend a firm's efforts. Firms that innovate overseas also face "liabilities of foreignness" (Zaheer, 1995), including "additional costs arising from the unfamiliarity of the environment, from cultural, political, and economic differences, and from the need for coordination across geographic distance" (p. 341). In terms of patenting, Kumar and Zaheer (2018) have found that foreign firms spend more time and effort obtaining patent approvals than domestic firms—another hurdle for the would-be cross-border innovators.

Government support helps mitigate such difficulties (Kang & Park, 2012). With supporting policies such as China's Innovative City Policy, governments provide slack resources, including not only financial stimulus and loans but also "economic benefits arising from the accumulation of skilled labor, specialized suppliers, and knowledge in the location" (Devarakonda et al., 2018, p. 1112). These resources buffer firms against shortages of funds and help them conduct experiments, absorb failures, form collaborations, and facilitate product/service innovations (Chan et al., 2021; Lu et al., 2014; Nohria & Gulati, 1996). The clustering of related firms, suppliers, and service providers within "innovative cities" creates positive externalities for the focal firm and influences its ability to conduct R&D and other value chain activities in the global market (Devarakonda et al., 2018; Massimino et al., 2017).

To innovate in a more advanced and competitive market like the United States, emerging-market firms benefit greatly from slack resources obtained through government support. Compared to firms in developed countries, emerging-market firms have fewer resources and/or "firm-specific advantages" (Choo et al., 2020; Gaimon & Ramachandran, 2021), but government support provides economic benefits, increases firms' access to a larger pool of external knowledge from related industries, boosts innovation incentives via social influence, and fosters a competitive environment for innovation. Tan and Peng (2003) examined slack resources in Chinese firms and showed that slack does not inhibit their efficiency or performance. Slack resources, instead, help firms survive environmental changes and supply chain disruptions (Hendricks et al., 2009) endemic to global competition and cross-border innovation. Guided by innovation policies, firms are less likely to misuse slack resources (Jensen, 1993). Taken together, emerging-market firms can fruitfully take advantage of government support to better mitigate uncertainties in cross-border innovation and achieve higher inventive productivity in their global operations. Therefore, we propose the following hypothesis.

Hypothesis 1 (H1) *Firms increase cross-border innovation under government support.*

2.3 | Role of R&D capacity in cross-border innovation

Although slack resources are generally believed to facilitate product/service innovations (e.g., Dewar & Dutton, 1986; Hitt et al., 2016), the magnitude of this effect varies across individual firms (Guo et al., 2016). Some studies (e.g., Aerts & Schmidt, 2008) report that local government support can help increase firms' capability and enable them to compete in the global market, whereas others note that emerging-market firms may lack their own capabilities, so government support can often be "ineffective and misleading" (Rugman & Li, 2007). These mixed effects of slack resources are well recognized in the literature. As Tan and Peng (2003) pointed out, slack can provide organizations with opportunities to develop new products and enter new markets, but it can also be a financial burden that breeds inefficiency and inhibits performance. Therefore, "a contingency perspective" is needed "to specify the nature of slack when discussing its impact on firm performance" (Tan & Peng, 2003, p. 1249).

A plausible contingent factor for explaining the mixed findings is a firm's "absorptive capacity." *Absorptive capacity* refers to a firm's ability to "recognize the value of new, external information, assimilate it, and apply it to commercial ends" (Cohen & Levinthal, 1990; Zahra & George, 2002). It can affect cross-border innovation from two aspects. On the one hand, firms with a higher level of absorptive capacity can more efficiently utilize slack resources (Leiponen & Helfat, 2010). They can absorb "unabsorbed slack resources" (e.g., financial stimulus, loans, and tax benefits) (Voss et al., 2008) and transform them into "absorbed slack" (e.g., nonliquid resources) to better deal with managerial complexity and environmental risks (Wiengarten et al., 2017). On the other hand, it may be that firms with already-stronger absorptive capacity are better bets for government support, as they can demonstrate their better use of slack resources and may be more likely to receive (a possibly larger allocation of) such support. Both effects signal the plausibility of absorptive capacity as a moderating factor.

A firm's R&D capacity consists of a set of "know-hows" that enable the firm to respond to market conditions, identify relevant new technologies, develop plans to exploit them, and acquire the necessary resources to do so (Cirera et al., 2021). A firm's R&D capacity reflects its absorptive capacity; therefore, it also plays a key moderating role in cross-border innovation. To innovate overseas, emerging-market firms must endure global shocks that require them to accelerate technology diffusion and enable more sophisticated forms of innovation and discovery (Cirera et al., 2021). Specifically, firms face pressures to "re-examine their own 'global' value chains, re-evaluate their cross-border investments strategies, re-assess their innovation and technology flows, and re-consider their strategic partnerships from a new perspective" (Petricevic & Teece, 2019, p. 1488). By better absorbing slack resources through government support, firms can, in turn, better address the competitive pressure with both domestic and foreign firms. They can also more

effectively handle the uncertainty of innovation in the foreign market and translate resources into innovative products and processes (Gann & Salter, 2000). In the absence of high R&D capacity, a firm can still benefit from access to government support but will be limited in its ability to leverage the benefits to improve its innovation performance (Kang & Lee, 2008). Therefore, we argue a synergistic effect between a firm's R&D capacity and government support on the firm's cross-border innovation.

Hypothesis 2 (H2) *Firms' R&D capacity amplifies the effect of government support on cross-border innovation.*

2.4 | Rebalancing innovation portfolio under government support

Research has shown that slack resources can increase firms' innovation capabilities (e.g., Nohria & Gulati, 1996), but less is known about how slack resources change innovation portfolios, which consist of two broad types of innovation—*exploration* and *exploitation* (March, 1991). Exploration refers, as noted above, to “radical innovations” that have “revolutionary changes in technology and clear departures from existing practices” (Chao & Kavadias, 2008; Dewar & Dutton, 1986, p. 1422), whereas exploitation refers to “incremental innovations” that have “minor improvement and simple adjustments to current technologies” (Chao & Kavadias, 2008; Dewar & Dutton, 1986, p. 1423). Scholars believe that a firm should maintain a *balanced* composition of radical and incremental innovation in its innovation portfolio (e.g., Chandrasekaran et al., 2012, 2016; Ladas et al., 2022; Si et al., 2022). By doing so, firms can better address uncertainty (Patel et al., 2012), overcome the productivity dilemma (Adler et al., 2009), promote social learning (Ladas et al., 2022), and achieve superior performance to firms emphasizing one strategy at the expense of the other (O'Reilly & Tushman, 2008).

However, exploration and exploitation have been understood as contradictory forces (e.g., March, 1991) with paradoxical requirements, competing for organizational resources (March, 1991). Some research argues that firms fail to balance exploration and exploitation when they lack resources (Choo et al., 2020; He & Wong, 2004; Jayanthi & Sinha, 1998). Particularly, Choo et al. (2020) conceptualize resource constraint as an even larger challenge to cross-border innovation. To address resource constraint, researchers have proposed approaches including resource allocation strategies (Hutchison-Krupat & Kavadias, 2015) and new innovation portfolio management frameworks (Si et al., 2022).

We argue that government support alleviates resource constraint and helps firms achieve balanced innovation portfolios. In other words, government provides slack resources that enable firms to strengthen the type of innovation that they previously did not emphasize. For instance, firms can utilize more financial slack to achieve higher exploration, or

they can adopt more operational slack to increase exploitation (Voss et al., 2008). Slack resources facilitate firms' pursuit of *ambidexterity*, so they can balance between short-term and long-term performance and survive or thrive amid international competition (Chao & Kavadias, 2008). Below, we develop two hypotheses and delineate *how* firms balance (or rebalance) radical and incremental innovation under government support.

First, government support enables exploitation-oriented firms (i.e., firms that have emphasized exploitation over exploration) to adopt more exploration in cross-border innovation. With slack resources, these firms are more willing to experiment and take risks (Steensma & Corley, 2001). That is, firms tend to invest resources in search of new competencies instead of enhancing their current strategic positions in a risky, uncertain, and competitive environment like cross-border innovation (Voss et al., 2008). Intensified competitive pressures in the global market prompt firms to seek extra, external resources from partners (Beckman et al., 2004) and call for exploration that can drive change and nurture new sources of competitive advantage (Levinthal & March, 1993). For “firms ... overly focused on incremental innovation and paid too little attention to long-term, radical innovation” (Si et al., 2022, p. 3), government support could change the status quo. Therefore, we propose the following hypothesis:

Hypothesis 3A (H3A) *With government support, exploitation-oriented firms will reduce the proportion of exploitation relative to exploration in their cross-border innovation portfolios.*

Second, for exploration-oriented firms (i.e., firms that have emphasized exploration over exploitation), government support can increase overseas exploitation. Note, for instance, that exploration-oriented firms can exhibit promising long-term prosperity but are perhaps concerned with short-term success (Uotila et al., 2009). They need to demonstrate their capability in efficiently converting innovation investments to profitable outcomes. In such scenarios, they would emphasize managing processes that transform unabsorbed slacks (e.g., financial slack) into absorbed slacks (e.g., human resource slack and operational slack) (Argilés-Bosch et al., 2016), because absorbed slacks improve operational efficiency. This supports existing projects in an exploitative manner and promotes incremental innovation (Voss et al., 2008). In fact, researchers have found that firms gain market success when they increase process management practices and redefine an existing product's functionality in new market applications (Benner & Tushman, 2003; Christensen, 1992). Taken together, we propose the following hypothesis:

Hypothesis 3B (H3B) *With government support, exploration-oriented firms will increase the proportion of exploitation relative to exploration in their cross-border innovation portfolios.*

3 | RESEARCH METHOD

3.1 | Policy enactment as quasi-experiment

China is one of the many countries that have enacted/are enacting policies to support regional innovation. Its policy intervention was instituted by the Chinese government at the national level, with city governments implementing the policy with upper-level guidance. Shenzhen was the first “innovative city” in 2009 (Shenzhen Government, 2009), and it was followed by 60 other cities on a rolling basis (MOST, 2010; MOST & NDRC, 2016). Since 2017, the most recent year in our dataset, 61 cities in China have implemented the policy at the city level (see Supporting Information Appendix B for a complete list). Firms located in an innovative city enjoy financial and institutional benefits provided by the government to enhance competitiveness in both domestic and international markets (see Supporting Information Appendix A for key policy instruments).

The United States and China are the two biggest economies in the world. China provides an important context and has received extensive attention in OM research (see, e.g., Cachon et al., 2020; Chen et al., 2009). Focusing on China, a single country, allows us to reduce the threat of heterogeneity across countries and avoid skeptical findings (Joglekar et al., 2016). This choice is supported by past findings that, in general, context-specific and evidence-driven studies have “yielded innovative theories and models for best in class performance” in the OM field and “will continue to do so as the field grows and matures” (Joglekar et al., 2016, p. 1994).

3.2 | Data and variables

We create our sample using policy data on innovative cities in China and patent data from the U.S. Patent and Trademark Office (USPTO). Patents are territorial in nature, and they must be filed in each country where protection is sought. Researchers assert that “patents from different patent authorities across the world are not comparable to each other” (Berry, 2014, p. 875). Thus, we follow the literature and use U.S. patents to create a representative sample, because “the United States offers the most attractive market for inventors ..., and inventors from China proactively file high-quality patents ... Thus, patents granted by the USPTO to Chinese inventors well represent China’s innovation capacity” (Zheng & Wang, 2020, pp. 2242–2243).

The panel matches the policy data to the USPTO data. The policy data contain the cities and the year when the cities implemented the policy (see Supporting Information Appendix B). The USPTO data contain patent applications in the United States made by firms and individuals from all over the world. The USPTO data contain the following information regarding patent application: inventors, assignees (i.e., firms in our study), city and country of each assignee (firm), application year, patent claims, citations, patent classes and sections, and so on. Because a firm may be recorded under

different names and locations in the raw patent data, we rely on the PatentsView platform, which standardizes the firm names and locations across all patents, reducing ambiguity and increasing the usability of the USPTO data.

To create the panel, we extracted all patents related to Chinese firms; this involved selecting patents awarded to firms in the category of “foreign corporations” and located in China. One key step involved integrating the policy data with the patent data by the city location of the firm. To do so, we first standardized the location information for firms in each patent via (1) filling in missing city information, (2) cleaning the city names (e.g., correcting misspellings), (3) transforming the names to geostandardized spatial data, (4) performing manual checking, and (5) consolidating the data and assigning a unique, correct city to a firm (by detecting the city that is most reported on patents for a given firm). We then cross-linked the patent data to the policy data by city. We select a time range of 2005–2017, to cover the entire period of the innovative city policy and make it long enough for the DiD analysis. The resulting panel has 7785 firms with 13,077 firm-year observations.

3.2.1 | Dependent variables

Our main analysis examines two dependent variables—*cross-border innovation* and *proportion of exploitation*—to test Hypotheses 1 and 2 and Hypotheses 3A and 3B, respectively. The first dependent variable *cross-border innovation* is operationalized as the number of patents granted by the USPTO for a firm in a year. We believe this is among the most appropriate operationalizations because the approval of a patent by the foreign regulatory institution is an acknowledgment of the novelty and inventiveness of the patent and ultimately reflects a firm’s successful patenting and innovation process overseas (Chan et al., 2021; Gittelman, 2008; Griliches et al., 1987; Kavadias & Ulrich, 2020). A number of studies have used patents to capture firms’ innovation activities, such as Chan et al. (2021) and Flammer and Kacperczyk (2016). To link the patent measure to the time when innovation activity was undertaken, we focus on the patent application year rather than the approval year, as there may be arbitrarily long lags between filing and approval dates for patents (Benner & Tushman, 2002; Flammer & Kacperczyk, 2016). The “year” throughout the paper refers to the application year of the granted patent unless otherwise specified.

The second dependent variable is the *proportion of exploitation* with respect to a firm in a year. Following Benner and Tushman (2002), we first examine the level of exploitation for each patent, which is referred to as the extent to which the patent is anchored on a firm’s existing knowledge. We regard repeat citations (i.e., patents that the firm had previously cited) and self-citations (i.e., the firm’s own previous patents) as existing knowledge. For each patent, the ratio between the citations belonging to existing knowledge and the total citations is its level of exploitation. Next, we average the level of exploitation for firm-year combinations to

obtain a firm's proportion of exploitation in that year. This metric captures the relative distribution of exploration and exploitation in a firm's innovation portfolio.

3.2.2 | Independent variables

We use three critical independent variables to investigate the role of government support in firms' cross-border innovation. First, *government support* is a binary variable indicating whether the firm's city was an innovative city in a given year. Specifically, the variable takes the value of 1, when the innovative city policy was enacted in the year, and for all the subsequent years for a city ("treatment group"), and 0 otherwise ("control group"). This measure is a commonly used DiD estimator (e.g., Flammer & Kacperczyk, 2016), and it can help identify the treatment effect of the government support through the DiD technique. Here, we examine the change in firms' cross-border innovation before and after receiving government support (first difference) between the treatment and control groups (second difference).

R&D capacity is operationalized as the number of inventors in a firm within a year (Hanaki et al., 2010). We obtain the measure by counting unique inventors across all patents of the firm by year. By focusing on inventors, we emphasize R&D capacity as related to human/talent capital invested by the firm. That is, this measure enables us to capture firms' absorptive capacity. Our measure of R&D capacity as absorptive capacity is derived from the commonly used measures like human capital and employee expertise (Zahra & George, 2002).

We create two dummies—*initial exploitation* and *initial exploration*—to capture firms' initial innovation strategy. Following Benner and Tushman (2002), we use the following criteria to define a firm's initial innovation strategy. First, for a year prior to the policy enactment in a city, if the firm's proportion of exploitation is greater than 0.6, it is determined to have employed the exploitation strategy in that year (*initial exploitation* = 1 and *initial exploration* = 0). If the proportion of exploitation is less than 0.4, that firm is determined to have adopted the exploration strategy in that year (*initial exploitation* = 0 and *initial exploration* = 1). Where a firm adopts a balanced strategy, both variables are equal to 0. Second, for a year into (or subsequent to) the policy enactment, we examine whether these variables remain unchanged from the most recent year before the policy enactment. We note that some firms are only present in our sample after the policy enactment. Due to a lack of information, we are unable to assign appropriate initial innovation strategy to those firms. Therefore, we remove those firms in the related analysis, leading to 2938 firm-year combinations for the variables.

The analysis also includes the following control variables. *Collaboration frequency* is a firm's patent co-filing activities with other firms in a year. Recent studies find that firms may collaborate with other firms for joint business relationships to strengthen knowledge pooling, cross-fertilization, and innovation performance (Chu et al., 2019; K. Li et al., 2019).

Therefore, we control for the effect of collaboration intensity on innovation by including *collaboration frequency*. This variable accounts for the fact that a patent can be filed by more than one firm. We obtain this variable by summing the number of other firms across patents for a focal firm within 1 year.

To tease out the influence of the quality and potential economic impact of a firm's innovation on patent outcomes, we control for the *number of claims*. This variable is the average number of claims across patents of a firm within a year. A patent has one or more claims that define what subject matter is protected by the patent. These claims reflect the nature and scope of the patent invention (scientific conceptualization vs. empirical discoveries and clarification), as well as the market value in need of legal protection and the impact of the following inventions. A patent with more claims will more likely withstand legal challenges. The *number of claims* provides a proxy for the overall legal strength of a firm's patents (Huang & Li, 2019).

The innovation outcome may also be affected by a firm's learning pattern and knowledge-sourcing decisions. Firms can choose to exploit existing knowledge in their home country or significantly advance technological development and "expand the firm's knowledge base beyond that existing in their home country" (Almeida, 1996, p. 157). We control for *home-country citation*, a variable that measures the extent to which a firm's patents cite other Chinese firms' patents (i.e., Chinese citations). We obtain the proportion of Chinese citations for a patent and then average the proportions across patents at the firm-year level. The variable reflects the extent to which the firm learns from domestic (Chinese) knowledge.

Because technological opportunities, development intensity, and patenting speed can vary across technological fields, we control for a firm's scope and focus of technological domains using the following two variables (Choudhury & Haas, 2018; Miao et al., 2021). First, we use the data of cooperative patent classification from USPTO to classify patents into sections indicating the subject to which the invention broadly relates. The variable of *number of sections* is measured by the number of unique sections across patents for a firm in a year to capture the broadness of the firm's innovation scope (Lerner, 1994). Second, we obtain the *primary section* of a firm by identifying the patent section with the highest frequency across all the firm's patents in a year. This variable captures the industry or the area in which the firm primarily innovates.

The analysis also includes each firm and each application year as the fixed effects in the empirical model. These variables account for idiosyncrasy and heterogeneity across firms and time.

3.3 | Empirical model

We adopt the generalized DiD technique to examine the effect of government support on cross-border innovation. In our context, a key challenge in identification is the

potential endogenous selection of the treatment cities. To mitigate this concern, we employ a matching method together with DiD (i.e., Matching-DiD) as our main analysis. We discuss endogeneity issues further in Section 3.3.3.

3.3.1 | Matching

It is challenging to draw a causal inference from observational data, even when the policy intervention is plausibly exogenous. Therefore, we employ the matching method to improve comparability between treatment and control groups and make better causal inferences. We conduct propensity score matching (PSM) at the city level, then, as a robustness check, perform additional matching techniques (e.g., coarsened exact matching (CEM) and rolling matching (RM)) at other levels (e.g., the firm level).

To conduct the PSM at the city level, we collect extra data based on Kou and Liu (2017). These data contain city statistics including city economic, investment, and general innovation performance. We use the following variables in the data for matching: population, gross domestic product (GDP), gross industry output, foreign investment, fixed asset investment, public finance expenditure, scientific research expense, educational expense, percentage of secondary industry employees, percentage of tertiary industry employees, and city innovation index. We also include geographical information so as to search for geographically close pairs of cities. We adopt one-to-one matching with the nearest neighbor measured by the Mahalanobis distance. Specifically, for each city in the treatment group in a year, we examine all cities in the control group in that year and select the one with the shortest Mahalanobis distance, considering all the city-level variables 4 years prior to the policy (Flammer & Kacperczyk, 2016).

Table 1 presents our balance diagnostics, showing an improved balance in covariates and overall propensity scores (“Distance (PSM)”). We compare the cities in the treatment group to the cities in the control group across all years. The differences in the distance and other covariates between the control and treatment groups are reduced and become insignificant in postmatching, indicating a high and increased level of comparability between treatment and control groups. Figure 1 also demonstrates this effect, as the density curves for Mahalanobis distance become much closer between the treatment and control groups after matching. The final matched sample contains 50 pairs of cities resulting in 10,993 firm-year combinations.

3.3.2 | Difference-in-differences

Because the time (year) of the innovative city policy (“treatments”) is different across cities (see Table A1 in Supporting Information Appendix B), we are analyzing a *staggered* quasi-experimental setting. The generalized DiD model as

suggested by Wooldridge (2019, Chapter 14-4) is most suitable for this kind of setting when multiple treatment groups receive the treatment at different time points. Specifically, then, we estimate the following regression:

$$\begin{aligned} & \text{Cross-Border Innovation}_{ict} \\ &= \alpha_i + \alpha_t + \beta_1 \cdot \text{Gov Support}_{ct} + \gamma' X_{ict} + \varepsilon_{ict}, \quad (1) \end{aligned}$$

where i indexes firms; t indexes years; α_i and α_t are firm and year fixed effects, respectively. The dependent variable of interest is *Cross-Border Innovation*, the number of U.S. patents. *Gov Support* is the post-treatment dummy for the enactment of the innovative city policy, as earlier described. X is the vector of control variables. ε is the error term. The analysis uses a Poisson regression for the model because the dependent variable is a count measure. To account for serial correlation of the error term, we cluster standard errors at the city level (Flammer & Kacperczyk, 2016). The coefficient of interest is β_1 , which measures the effect of government support on cross-border innovation. Hypothesis 1 predicts that β_1 should be positive and significant.

To examine the moderating role of a firm’s R&D capacity, we add an interaction term to the regression. That is,

$$\begin{aligned} & \text{Cross-Border Innovation}_{ict} \\ &= \alpha_i + \alpha_t + \beta_2 \cdot \text{Gov Support}_{ct} \times \text{R\&D Capacity}_{ict} \\ &+ \gamma' X_{ict} + \varepsilon_{ict}, \quad (2) \end{aligned}$$

where the symbols carry the same meaning, except that the coefficient of interest becomes β_2 , which measures how *R&D Capacity* moderates the effect of government support on cross-border innovation. Hypothesis 2 predicts that β_2 should be positive and significant.

To examine whether firms adjust their portfolio of exploratory and exploitative innovations after receiving the government support, we conduct subgroup analysis. Based on firms’ *initial exploitation* and *initial exploration* values, we classify the firms into initially exploiting, initially balanced, and initially exploring. For each group, we examine the firm’s overall proportion of exploitation to identify how the firm shifts its strategic focus between exploitation and exploration. The following model applies to any of the three groups:

$$\begin{aligned} \% \text{ Explt}_{ict} &= \alpha_i + \alpha_t + \beta_3 \cdot \text{Gov Support}_{ct} + \gamma' X_{ict} + \varepsilon_{ict}, \\ & \quad (3) \end{aligned}$$

where the symbols carry the same meaning as those in the previous models. However, the dependent variable becomes *% Explt*, a firm’s mean proportion of exploitation across all its patents in a year. Because the dependent variable is a ratio, we adopt the generalized linear regression with the logit link. The coefficient β_3 measures the extent to which government

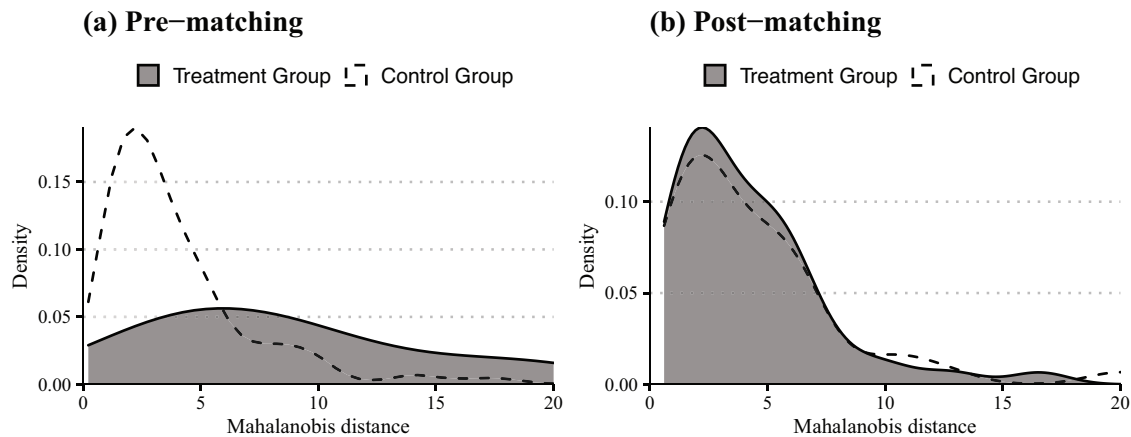
TABLE 1 Balance of matched cities

		Prematching Control	Postmatching		<i>t</i> -statistic	<i>p</i> -value
			Treatment	Control		
1	Distance (PSM)	0.15	0.59	0.32	0.33	0.93
2	Population (×10,000)	606.11	574.08	399.50	457.92	0.14
3	GDP ^a	14,611,165.02	20,996,169.91	16,829,672.11	16,560,080.46	0.88
4	Gross industrial output ^a	17,771,524.85	34,365,761.15	30,592,907.01	26,024,393.68	0.23
5	Foreign investment ^a	650,484.91	5,765,883.49	3,334,698.62	2,582,310.23	0.28
6	Fixed asset investment ^a	8,865,299.75	13,010,044.26	9,418,707.41	9,545,235.26	0.12
7	Public finance expenditure ^a	1,708,188.02	2,365,617.21	1,771,579.03	1,740,565.27	0.17
8	Scientific research expense ^a	23,145.62	29,187.33	25,821.06	24,818.11	0.17
9	Education expense ^a	301,887.73	338,083.94	304,847.12	294,114.01	0.28
10	Secondary industry employees (%)	48.50	48.24	47.85	46.61	0.46
11	Tertiary industry employee (%)	42.24	48.53	44.94	42.94	0.91
12	Innovation index	3.61	6.45	3.81	3.97	0.35

Note: For balance diagnostics, we apply propensity score matching on cities with yearly pooled data. The prematching data include 177 cities and 709 city-year observations. The postmatching data include 80 cities and 168 city-year observations.

The variables are log-transformed for matching. The matching is based on one-to-one nearest matching with the propensity score distance and a caliper of 0.05.

^aMatching variables are measured in the unit of 10,000 CNY.

**FIGURE 1** Density curves for Mahalanobis distance

support promotes the change in firms' cross-border innovation strategy. Hypothesis 3A predicts that, for the initially exploiting group, β_3 should be negative and significant, while Hypothesis 3B predicts that for the initially exploring group, β_3 should be positive and significant.

3.3.3 | Exogeneity of policy enactment to firms

Policy enactment is a critical intervention in this study. This subsection justifies the exogeneity of the policy enactment to the firms in the innovative cities, strengthening the validity of our identification strategy. That is, the enactment of the policy should not be driven by individual firms' decisions, strategies, or activities. We conduct both contextual and empirical examination in a stepwise manner as a verification.

As the first step, we search for qualitative evidence that would suggest that firms can affect and take advantage of policy enactment. The policy was approved by the Chinese government at the national level and later implemented by city governments. We search the CNKI database¹ but do not find evidence that suggests firms intentionally take actions such as lobbying to affect the implementation of the policy. We also examine the headquarters of the firms in our sample during 2005–2017, ruling out the possibility that firms might know about the future enactment of the policy beforehand and intentionally pursue resources by moving.

In the second step, we empirically mitigate the endogeneity concern by checking the parallel trends assumption of DiD. For brevity, we report empirical results and details in Supporting Information Appendix D and visualize the parallel trends assumption in Figure 2, which presents the mean counts of

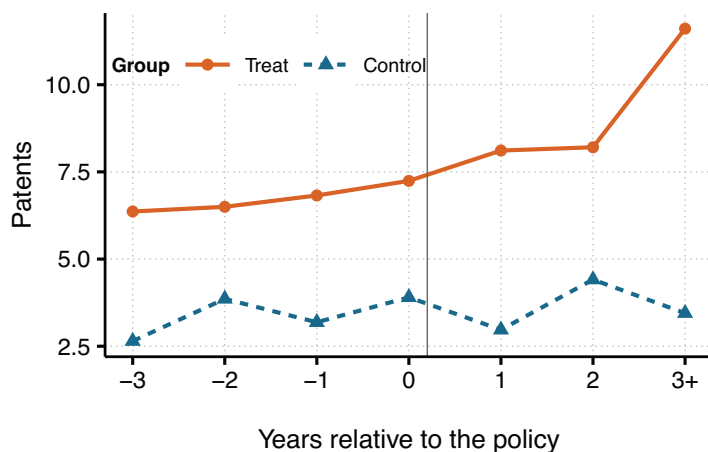


FIGURE 2 Parallel trends analysis. We consider firms that have 2 or more years of data. For the treatment group, year 0 indicates the year of the policy enactment in the city. For the control group, year 0 indicates the median year for a firm in the sample. [Color figure can be viewed at wileyonlinelibrary.com]

United States patents for the treatment and control groups and, together with the empirical results in Supporting Information Appendix D, demonstrates that the parallel trends assumption holds. Both the treatment and control groups have the same trend of filing foreign-country (U.S.) patents before the policy intervention, while there is a diverging trend of patenting after the government support.

The third step is to allay concerns about sample selection and sample heterogeneity that may affect both innovation and the treatment. Taking advantage of the staggered nature of the policy enactment, we re-estimate the specifications using only the eventually treated firms, which were first in the control group and only later in the treatment group. In other words, the control group consists exclusively of firms that are eventually treated (see Flammer and Kacperczyk, 2016, for a similar argument). The re-estimation results (see Section 4.3.2) remain consistent with the main results, indicating that sample selection or heterogeneity is not a concern.

4 | MAIN RESULTS AND ROBUSTNESS CHECKS

4.1 | Descriptive statistics

Table 2 shows the summary statistics and correlations of the numerical variables in the main analysis (Matching-DiD). It shows that *government support* exhibits a positive association with *cross-border innovation*. Considering firms' tendency for exploitation, *government support* is negatively related to the firms' *proportion of exploitation*. In addition, *government support* is positively related to *R&D capacity* and other control variables. Furthermore, *R&D capacity* has a strong positive association with *cross-border innovation*.

4.2 | Hypothesis testing results

4.2.1 | Government support and R&D capacity in cross-border innovation

Table 3 presents the DiD results that verify Hypotheses 1 and 2. We introduce the control variables in Model (1), the

main independent variables in Model (2), and the full model with the interaction term in Model (3). Models (2) and (3) correspond to our empirical specification as Equations (1) and (2).

Model (2) in Table 3 shows that *government support* has a positive and significant effect on *cross-border innovation* ($\beta_1 = 0.176, p < 0.01$), suggesting that government support leads to increased patents in the United States. Thus, Hypothesis 1 is supported.

Model (3) verifies the moderating effect of *R&D capacity*. The results show that *R&D capacity* has a significant and positive interaction effect with *government support* on *cross-border innovation* ($\beta_2 = 0.039, p < 0.05$), thus supporting Hypothesis 2. The results imply that the positive effect of *government support* on *cross-border innovation* is stronger for firms with a higher level of *R&D capacity*. Thus, compared to firms with lower R&D capacity, those with higher R&D capacity (higher absorptive capacity) can more easily identify, assimilate, and deploy external resources and opportunities—put simply, can more easily innovate.

Model (3) reports interesting results for control variables. For example, *collaboration frequency* has a positive significant effect on *cross-border innovation* ($\beta = 0.0735^{***}$), indicating that collaboration among firms helps improve innovation overseas. Meanwhile, both *number of claims* and *home-country citation* have negative significant effects on *cross-border innovation* ($\beta = -0.1305^{***}$ and $\beta = -0.1465^*$, respectively). These negative results seem to suggest that cross-border innovation decreases when firms pursue stronger legal protection or exploit more domestic knowledge for their innovation projects.

4.2.2 | Rebalancing the portfolio of cross-border innovation

We have hypothesized that government support can affect a firm's strategic focus on cross-border innovation and lead to a rebalancing of exploration and exploitation (Hypotheses 3A and 3B). With government support, firms can readjust their innovation portfolio to achieve superior performance, simultaneously exploring and exploiting innovation

TABLE 2 Summary statistics and correlations of numerical variables

Variables	Mean	SD	Median	Min	Max	1	2	3	4	5	6	7	8
1 Cross-Border innovation	5.55	42.79	1	1	1563	1							
2 Proportion of exploitation	0.70	0.42	1	0	1	−0.075***	1						
3 Government support	0.68	0.47	1	0	1	0.039***	−0.023**	1					
4 R&D capacity	9.51	53.52	3	1	1933	0.952***	−0.089***	0.042***	1				
5 Collaboration frequency	2.03	20.25	0	0	916	0.630***	−0.069***	0.029***	0.532***	1			
6 Number of claims	12.08	6.48	11	1	78	0.025**	−0.072***	0.032***	0.035***	0.020*	1		
7 Home-country citation	0.26	0.29	0.17	0	1	0.041***	0.067***	0.121***	0.057***	0.022*	0.021*	1	
8 Number of sections	1.88	1.09	2	1	9	0.338***	−0.181***	0.078***	0.341***	0.301***	0.053***	0.045***	1

Note: Pearson's paired test for variable correlations in the full sample. The number of observations for the full sample is 13,077.

Significance codes: ***0.001, **0.01, *0.05 (two tailed).

TABLE 3 Two-way fixed-effects poisson regression on cross-border innovation

Dependent variable: Model:	Cross-border innovation		
	(1)	(2)	(3)
<i>Independent variables</i>			
lg_Collaboration Freq	0.305*** (0.033)	0.075*** (0.018)	0.073*** (0.017)
lg_Number of claims	−0.197*** (0.059)	−0.134*** (0.020)	−0.130*** (0.021)
Home-Country Citation	0.162 (0.114)	−0.151* (0.071)	−0.146* (0.069)
lg_Number of Sections	1.271*** (0.195)	0.323*** (0.034)	0.318*** (0.034)
lg_R&D Capacity		1.005*** (0.063)	0.968*** (0.063)
Government Support		0.176** (0.064)	0.109 ² (0.065)
Gov Support × lg_R&D Cap			0.039* (0.019)
<i>Fixed effects</i>			
Assignee	Yes	Yes	Yes
Application year	Yes	Yes	Yes
Primary section	Yes	Yes	Yes
<i>Fit statistics</i>			
Log likelihood	−20,199.483	−15,754.377	−15,747.391
Observations	10,993	10,993	10,993
Pseudo- <i>R</i> -squared	0.845	0.871	0.871

Note: Standard errors (reported in parentheses) are clustered at the city level. The sample period is from 2005 to 2017. Independent variables are log-transformed to avoid skewness. *lg_R&D Capacity* is further scaled to avoid multicollinearity due to the interaction.

Significance codes: ***0.001, **0.01, *0.05, +0.1.

opportunities. Specifically, exploitation-oriented firms develop a larger share of exploratory innovation and vice versa. Table 4 presents the results pertaining to the empirical specification (Equation 3).

Table 4 presents the subgroup analysis. Model (1) in Table 4 shows the impact of government support on initially exploring firms. The coefficient of *government support* is positive and significant ($\beta_3 = 0.448, p < 0.001$), supporting Hypothesis 3B. Results imply that firms with initially high exploratory innovation increase their proportion of exploitation after receiving support from the government. Meanwhile, we find empirical evidence supporting Hypothesis 3A. Model (3) in Table 4 is for initially exploiting firms. The results show a negative and significant coefficient of *government support* ($\beta_3 = -0.555, p < 0.001$), suggesting that firms with initially high exploitative innovation reduce the proportion of exploitation after receiving support from the government. Overall, we have verified that government support promotes a rebalance in a firm's innovation portfolio.

4.3 | Robustness checks

We conduct a series of robustness checks to show that our findings are not driven by sample selection (Sections 4.3.1 and 4.3.2) or variable measures (Sections 4.3.3–4.3.5), then examine the influence of corporate structure (Section 4.3.6). We report the key points of the analyses and findings in the main text but refer readers to the appendices for the technical details.

4.3.1 | Alternative matching methods

We have adopted matching at the city level with DiD as our main analysis. We conduct three alternative matching

TABLE 4 Two-way fixed-effects generalized linear model with logit link on the proportion of exploitation

Dependent variable: Model:	Proportion of exploitation		
	(1) Explr firms	(2) Bal firms	(3) Explt firms
<i>Independent variables</i>			
lg_Collaboration Freq	0.006 (0.018)	0.010 (0.036)	0.032 ⁺ (0.016)
lg_Number of Claims	−0.038 (0.036)	−0.024 (0.061)	−0.012 (0.025)
Home-Country Citation	0.218* (0.083)	0.041 (0.174)	0.135 ⁺ (0.076)
lg_Number of Sections	−0.042 (0.048)	0.015 (0.063)	0.027 (0.026)
lg_R&D Capacity	−0.001 (0.033)	−0.061 (0.051)	−0.040** (0.015)
Government Support	0.448*** (0.067)	0.195* (0.090)	−0.555*** (0.063)
<i>Fixed effects</i>			
Assignee	Yes	Yes	Yes
Application year	Yes	Yes	Yes
Primary section	Yes	Yes	Yes
<i>Fit statistics</i>			
Log likelihood	−236.317	12.886	270.263
Observations	1037	238	1635
R-squared	0.336	0.457	0.719

Note: Standard errors (in parentheses) are clustered at the city level. Sample period: 2005–2017. Independent variables are log-transformed to avoid skewness. Significance codes: ***0.001, **0.01, *0.05, +0.1.

techniques to strengthen the robustness of our findings. The methods include (1) CEM at the city level, (2) RM at the city level, and (3) PSM at the firm level. Supporting Information Appendix E1 describes the matching details and presents the results. We find that the new results are consistent with the main findings.

4.3.2 | Alternative samples: Full sample and a sample without never-treated firms

This subsection re-estimates the DiD specification using (1) the full sample and (2) the sample with only eventually treated firms. The full sample includes all the firm-year observations without matching, while the smaller sample excludes the firms that are never treated to mitigate the effect of unobserved differences of firms between the treatment and control groups on the policy and innovation (see Section 3.3.3 for discussion). In sum, Supporting Information Appendix E2 shows the two sets of re-estimation results, which remain consistent with the main analysis results, indicating that

our identification strategy is not affected by sample selection.

4.3.3 | Alternative measures and mechanisms for cross-border innovation

This subsection examines eight alternative measures for the dependent variable (DV), *cross-border innovation*. These measures help clarify potential alternative mechanisms for cross-border innovation.

1. *Considering a patent's priority.* The priority of a patent is a critical piece of information that links a later-filed patent application to an earlier-filed one. Therefore, it may imply the novelty of the patent and indicate whether a firm applies for patents in the United States merely to obtain the protection of their products in the U.S. market. We collect the priority information from PatentsView, which records the earlier patent filing in a non-U.S. country and the associated claimed priority for a granted patent. We examine three extra DVs: (1) Alt.DV #1: A firm's U.S. patents excluding those with long priority history (i.e., the earliest priority date is 3/2/1 years earlier than the application date), (2) Alt.DV #2: recounting the number of patents by replacing the application date with the earliest priority date, and (3) Alt.DV #3: a firm's U.S. patents with an additional control variable of the average duration between the application date and the earliest priority date. Supporting Information Appendix E3 shows that these analyses yield consistent results.
2. *Considering a patent's classes.* Patent classes describe general areas of technology embedded in patents. Each patent has at least one but may have multiple documented classes. Comparing a firm's patent classes over time, we can quantify the extent to which the firm is combining existing technologies or creating new technologies. Following Wu et al. (2019), we characterize a firm's innovation nature based on the combinations of patent classes (i.e., United States Patent Classification (USPC) patent classes). Developing measures from the pairs in the combinations of patent classes for a firm, we examine two more DVs: (1) Alt.DV #4: the number of pairs that are new and novel to the world (i.e., the combination pairs of technologies that have not been explored before), and (2) Alt.DV #5: the total number of combinations for a comparison. Supporting Information Appendix E4 shows that the novelty or quantity in combinations of patent classes is consistent with our findings about overall patent outcome (i.e., the total number of patents).
3. *Considering the location of inventors.* There are potentially two underlying mechanisms behind a U.S. patent. First, the inventive activities can be domestic (e.g., firms rely on local inventors to develop the technologies), and firms obtain protection in the U.S. patent office. Second, the inventive activities can be cross-border (e.g.,

Chinese firms engage foreign inventors or move employees to other countries). Although the outcomes are similar (i.e., more patents), the underlying activities and mechanisms can be fundamentally different. To reduce the concern that engaging foreign inventors may affect our results, we collect data about the inventors' location from PatentsView. We examine two more DVs: (1) Alt.DV #6: a firm's U.S. patents excluding those with U.S. inventors and (2) Alt.DV #7: a firm's U.S. patents excluding those with non-Chinese inventors. We add a control variable—the average percentage of Chinese inventors—by averaging the proportions of Chinese inventors across patents for a firm-year combination. Supporting Information Appendix E5 shows that the results are largely consistent with the main analysis results.

Note, however, that even with government support, the first mechanism (i.e., Chinese inventors creating U.S. patents) still accounts for a larger portion (~95% based on the number of observations) of cross-border innovation than does the second (i.e., Chinese firms engaging U.S. inventors). It lends support to the notion that cross-border innovation is quite risky.

4. *Filling zeros for years without patents.* This robustness check considers the years in which a firm does not apply for patents. In this case, we fill in zeros for the firm within the minimum and maximum of years of presence. This creates a lot of 0 values in the DV (Alt.DV #8) but alleviates the sample selection problem. Supporting Information Appendix E6 shows that our results remain consistent.

4.3.4 | Alternative measures for R&D capacity

In this subsection, we adopt three alternative measures for R&D capacity. First, we remeasure R&D capacity as the number of unique inventors over years $t-1$ to $t-5$, to alleviate the concerns that the original measure (1) may undercount unproductive inventors who file patents relatively infrequently and (2) is mechanically correlated with the DV (as an example, no inventors imply no patents). We re-estimate the DiD model using the main sample.

The second and third measures use R&D expenditure divided by total assets and total revenue, respectively. We re-estimate the DiD model with firm size (the number of employees) as an additional control. However, due to the limited financial information available for private firms in China, we examine a reduced sample of publicly traded Chinese firms listed in the U.S. stock market for validation purposes. We merge our sample with COMPUSTAT.

Supporting Information Appendix E7 presents more details. The results show positive and significant interaction effects between the government support and R&D capacity, further reinforcing the finding that a firm's R&D capacity amplifies the effect of government support on its cross-border innovation.

4.3.5 | Alternative measures for exploitation and exploration

In this subsection, we adopt four alternative measures for the three variables related to *initial exploitation*, *initial exploration*, and the second DV—*proportion of exploitation*.

1. *Using quantile-based cutoffs.* Instead of using 0.4 and 0.6, the new measure uses 0.2 and 0.533 as the cutoff to create *initial exploration* and *initial exploitation*, based on the 25% and 75% quantiles, respectively.
2. *Replacing initial exploration and initial exploitation with "initial proportion of exploitation."* Other than the subgroup analysis, this robustness check creates a continuous variable "initial proportion of exploitation" that essentially integrates and replaces the subgroups *initial exploration* and *initial exploitation*. We add an interaction term of "initial proportion of exploitation" and *government support* in Equation (3) as a re-examination. The results support H3A and H3B by showing that initially exploratory firms become more exploitative (coefficient = 0.357***; see Table E8 column (4) in Supporting Information Appendix E8), whereas initially exploitative firms reduce the share of their exploitative activities (coefficient = 0.357 – 0.958 = –0.601***). Moreover, by visualizing the policy effect separately for each group in Figure 3, we provide striking evidence of firms pursuing ambidexterity by rebalancing exploration and exploitation after an infusion of government support.
3. *Re-operationalizing existing knowledge using citation time window.* The main analysis follows Benner and Tushman (2002) and considers all the repeat- and self-citations that have ever appeared in the patents as a firm's existing knowledge. To account for the fact that the set of patents that the firm has previously cited gets bigger over time, we adopt a time window when we operationalize existing knowledge. That is, we filter for citations within 10 years from the date of the patent application; to avoid inflation, citations earlier than 10 years are not considered existing knowledge.
4. *Re-operationalizing existing knowledge using the "geo-location base."* Considering that frequent interactions between co-locating firms lead to knowledge flow, we extend a firm's existing knowledge from repeat- and self-citations to citations in the "geo-location base" of a firm, defined as all the locations (cities) of a firm's citations within 5 years of the patent application. We measure exploitation as the ratio between the number of repeated cities and the total number of cities in the citations.

Supporting Information Appendix E8 presents the results. All new results support H3A and H3B, confirming the robustness of our findings.

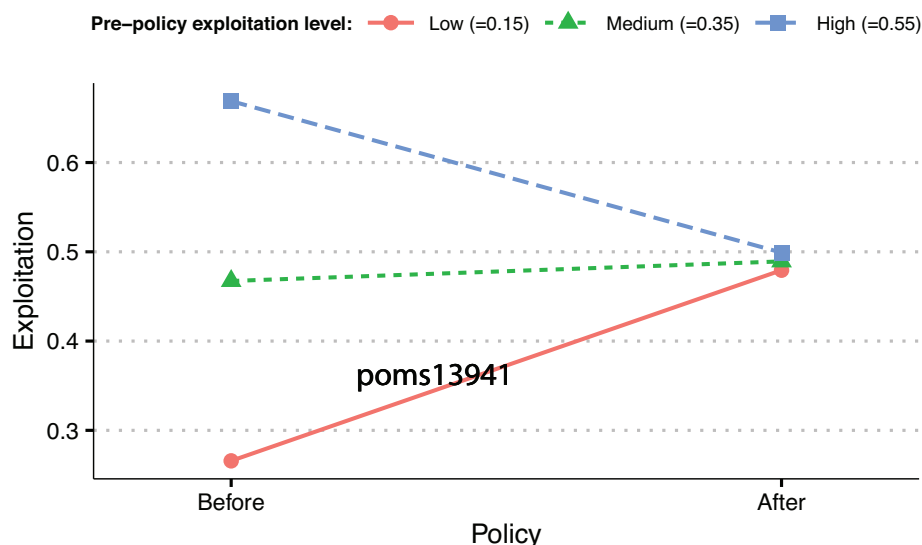


FIGURE 3 Interaction plots between policy and initial exploitation. This conditional effect plot shows the change in exploitation for the three subgroups using quartiles before and after the policy. The estimation is based on the model with all other variables at their mean values. [Color figure can be viewed at wileyonlinelibrary.com]

4.3.6 | Other factors and effects: Corporate structure

This subsection considers the situation in which firms in different locations belong to the same parent company. Because the original data from PatentsView (the December 2019 version) less clearly showed corporate structures, we manually searched for potentially related firms (assign a new dummy “multi-city” = 1) and analyzed how corporate structure could influence government support. We conduct two sets of analyses. First, we interact “multi-city” with *government support* and *R&D capacity*. Second, we include “multi-city” as the fixed effect to account for potential heterogeneity between these two types of firms. Supporting Information Appendix E9 shows that the results remain consistent with the main findings.

4.4 | Post hoc analysis on domestic and global patents

The main analysis focuses on Chinese firms’ U.S. patents. In this subsection, we conduct post hoc analysis of their domestic (i.e., Chinese) and global patents. We explore how the policy affects patent filings in other countries around the world.

First, we turn to China’s patent office for firms’ domestic patent information. We first identify top Chinese firms active in U.S. patenting (i.e., top 300 in terms of the total number of patents). We then conduct two sets of analyses. In the first, we perform a similar DiD analysis using the number of domestic patents as the new DV. In the second, we use the original DV but include domestic patents as a control variable. Table E11 in Supporting Information Appendix E10 presents the

results, which support our finding in a broader sense; that is, the policy intervention improves firms’ patenting proxied by the records from both United States and Chinese patenting offices.

Next, we examine firms’ patents all around the world. Here we turn to the European Patent Office (EPO)’s Worldwide Statistical Patent Database (PATSTAT) to obtain The Organisation for Economic Co-operation and Development (OECD) Harmonised Applicant Names (HAN) database. This database cleans and develops harmonized names for patent applicants (firms and individuals) from EPO, USPTO, and the Patent Co-operation Treaty, hence enabling us to identify all global patents belonging to the same firm and cross-match firms across HAN and USPTO. We examine four more models. Columns (1) through (3) in Table E12 in Supporting Information Appendix E11 use a firm’s non-U.S. global patents, a firm’s total global patents (including those in the United States), and a firm’s U.S. and EPO patents as new DVs, respectively. Column (4) in Table E12 examines the original DV, but with non-U.S. patents as a control variable. Results remain largely consistent, supporting a high level of generalizability of our findings to global patents.

5 | DISCUSSION AND CONCLUSION

5.1 | Findings and theoretical contributions

Cross-border innovation is an indispensable component of a firm’s global operations, and government support plays a critical role in fostering it. Both emerging and advanced markets have implemented or are implementing innovation policies along these lines. Through the lens of Porter’s diamond model, we conceptualize government support as slack

resources, which are key in helping firms increase “resource capabilities” and manage their internal and external “knowledge value chain” for improved performance (Gaimon & Ramachandran, 2021).

We first clarified the mixed literature on government support by showing how government support promotes cross-border innovation. Next, we showed that government support makes firms with high levels of absorptive capacity stronger. This finding demonstrates the need to take a “contingency perspective to specify the nature of slack when discussing its impact on firm performance” (Tan & Peng, 2003, p. 1249). Finally, we examined the role of government support in a firm’s innovation portfolio overseas. In line with the ambidexterity argument, in which firms balance between exploration and exploitation, this study shows that firms rebalance their global innovation portfolios after receiving the government support.

This study contributes to the OM literature in several ways. First, it examines the role of public policy and government support in OM, a field with a long tradition of examining how government regulation or policy affects firms’ operations (e.g., Joglekar et al., 2016; Kickul et al., 2011; Krishnan & Loch, 2005). This study addresses the “need to examine operational decisions with policy considerations” (Joglekar et al., 2016, p. 1977). We conceptualize government support as slack resources and clarify how slack affects firms’ cross-border innovation. While slack is traditionally understood as an *internal* slack resource, we add a new perspective on *external* resources provided through government policy. Our study highlights that government can provide slack resources to firms, echoing the importance of using slacks to facilitate firms’ innovations (Hitt et al., 2016). Broadly, our study complements recent findings of the role of government in OM, including cyber security (S. Luo & Choi, 2022), vaccine development (Adbi et al., 2019), electric vehicle promotion (Yu et al., 2022), and reactions to pandemics (Sodhi & Knuckles, 2021; Xu et al., 2022).

Second, this study helps inform the mixed literature on the effects of slack resources on firm performance. The analysis shows a positive impact of government support on firms’ cross-border innovation, supporting the positive effect of slacks on global innovation. This finding supports the view that government support provides a critical external resource for firm innovation (Gaimon & Bailey, 2013; Gaimon et al., 2017; Peng et al., 2014) and echoes Kavadias and Hutchison-Krupat (2020), which asserts that firms’ innovation activities are subject to the external political environment and general ecosystem. In sum, government support has the potential to help firms strengthen their innovation processes and mitigate risk and uncertainty in innovation projects (Linderman et al., 2004).

Third, this study proffers absorptive capacity as one of the key explanations for the mixed literature regarding government support and slack resources. Specifically, absorptive capacity explains the important practical observation that much heterogeneity exists in the innovation performance of firms in developing East Asia, despite those firms receiving

government support (Cirera et al., 2021). As firms expand into the competitive global market, absorptive capacity helps address pressures to “re-examine their own ‘global’ value chains, re-evaluate their cross-border investments strategies, re-assess their innovation and technology flows, and reconsider their strategic partnerships from a new perspective” (Petricevic & Teece, 2019, p. 1488).

Fourth, this study shows that government support leads to a rebalance between exploration and exploitation in firms’ global innovation portfolios. This finding is in line with the idea of ambidexterity, which argues that when firms balance their exploration and exploitation, they enhance their long-term competitiveness (March, 1991; He & Wong, 2004; Patel et al., 2012). This study also informs the productivity dilemma, which examines how firms address the tension between exploration and exploitation (Adler et al., 2009). Our findings imply that firms can change the way they address the productivity dilemma (from an innovation perspective) after the policy intervention of extra slack resources. We believe that this provides implications for managing NPD portfolios (Chandrasekaran et al., 2012, 2016; Kavadias & Ulrich, 2020), wherein firms seek to balance exploration (radical) and exploitation (incremental) innovations.

5.2 | Practical implications

Our research findings have implications for both managers and policymakers. For managers, we highlight the positive effect of government support on firms’ innovation process. Thus, managers may need to pay attention to the external political and regulatory environment and integrate their internal practices with external dynamics to achieve better innovation performance. In particular, government support can facilitate firms’ innovation in the global market. Managers should assess their internal R&D capacity (e.g., talent capital) to better utilize available external resources. In addition, firms can use government support to make a strategic shift and better align their short-term goals (exploitation) with long-term goals (exploration).

For policymakers, this study confirms that government support helps firms’ cross-border innovation. Depending on the city’s development plan and global ambition, the government should decide whether to give out a “package” of supports to enhance firms’ and cities’ innovation capabilities. The benefits can be even larger in firms with larger talent pools of R&D employees. Therefore, policymakers can improve the efficacy of their policy implementations by helping firms recruit, attract, and retain new talents. Additionally, as firms can shift their innovation focus with government support, our findings suggest government takes the initiative to promote firms’ innovation in general and to foster balanced innovation portfolios in particular. Nonetheless, policymakers should be aware that this policy intervention makes already strong, innovative firms stronger. To maintain healthy market competition, policymakers may study and redesign policy so as to nurture firms with lower R&D capacity.

Although developing East Asia (e.g., China) has been “home to some leading innovative companies that might rival those in high-income countries, there remains a large technological divide between and within the region’s countries” (Cirera et al., 2021, p. 105). Not only China, but other national and regional governments should consider enacting innovation policies to support firms in advancing technological development. The policy support is critical because emerging-market firms face major constraints, uncertainties, and challenges in global markets. Therefore, as Cirera et al. (2021) suggested, innovation policy should be tailored to “not only support firms pushing the frontier but also those firms investing in catching up” (p. 105).

5.3 | Limitation and future research

There are several possible extensions of this study that future researchers might pursue. First, this study focuses on U.S. patents from Chinese firms. This is helpful in reducing the threat of heterogeneity in analysis, but it overlooks the innovative efforts enacted in other emerging markets such as India and Malaysia (Joglekar et al., 2016). Although context-specific and evidence-driven studies have yielded innovative theories and models for best-in-class performance (Joglekar et al., 2016), we advocate future research examining cross-border innovation in multicontext or other countries. Similarly, it is important for firms to extend the scope of their innovation beyond the U.S. market. Future research will fruitfully examine how government support enables emerging-market firms to penetrate global markets like Japan and Europe as well as fellow emerging markets.

Second, this study focuses on innovation performance as determined by patent filings. This captures the earlier stage of ideas and inventions for new products and processes. However, we have not examined the product and service development process nor the launching of products and services (i.e., later stages). It is critical to further understand how government support affects the NPD and project portfolio management in detail. More research is needed to fully understand how government support intervenes in firms’ internal (e.g., process management, design, R&D activities, team collaboration) and external operations (e.g., collaboration with partners in the local communities and in the global supply chains).

Third, the policy in our context can affect foreign as well as domestic firms in the treatment group (innovative cities). Examining this outcome would flesh out our research and make it possible to identify any spillover effect such that government support changes the interactions (e.g., NPD collaborations) between local and foreign firms.

Finally, the policy in this study is implemented at the city level. As other countries implement policies at the national level, it would benefit the literature to determine which level of policy is the more effective intervention. The DiD method can tease out the city-level policy effect and leave no country-level confounding effects, even if there were nation-

wide policies. Future research can examine a different setup that compares nationwide and local support in two or more (comparable) countries.

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ENDNOTE

¹ CNKI (<https://cnki.net/>) is an online publishing platform of China knowledge resources initiated by Tsinghua University and Tsinghua Tongfang Holding Group. It provides “over 90% of China knowledge resources, the widest in title and type coverage and deepest in year coverage in China” and “comprehensive coverage of journals, dissertations, newspapers, proceedings, yearbooks, reference works, encyclopedia, patents, standards, S&T achievements and laws and regulations” (<https://oversea.cnki.net/index/Support/en/Project.html>).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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